

Code

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY NOT_gate IS
    PORT (
        a : IN STD_LOGIC;
        y : OUT STD_LOGIC
    );
END NOT_gate;

ARCHITECTURE Behavioral OF NOT_gate IS
BEGIN
    y <= NOT a;
END Behavioral;
```

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
```

라이브러리 호출 및 std logic 사용

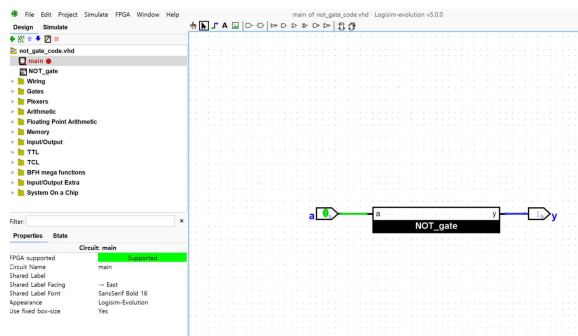
```
ENTITY NOT_gate IS
    PORT (
        a : IN STD_LOGIC;
        y : OUT STD_LOGIC
    );
END NOT_gate;
```

NOT_gate에 사용할 입력값과 출력값 지정

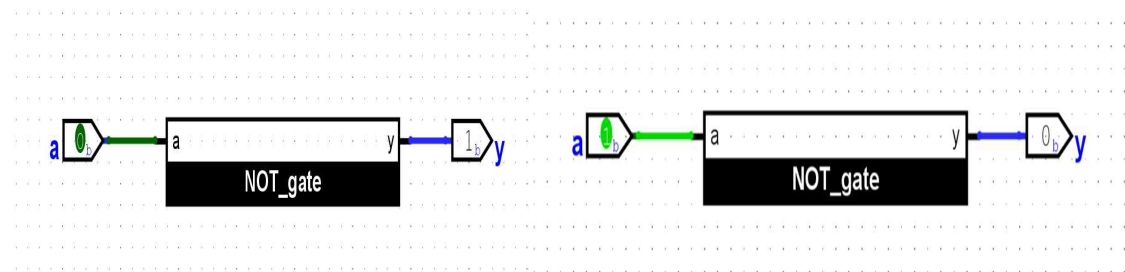
```
ARCHITECTURE Behavioral OF NOT_gate IS
BEGIN
    y <= NOT a;
END Behavioral;
```

NOT_gate의 출력값인 y에 NOT a 값 넣고 종료

Circuit



Result



a(0/1) 값이 NOT_gate를 통과해서 y(1/0) 값으로 변경됨